

Classification of Nutritional Deficiencies in Cabbage Leave Using Random Forest

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Abstract—Now a day agriculture is very important in India since it is a growing nation. But generally the crop production attained by farmers would be much below the optimal production. It is very important to correctly detecting and identifying the crop diseases to enhance the profit of the formers and the stakeholder. The main reason for the crop production gap is due to the lack of essential soil nutrients and irrigation in the agricultural farms. To escalate the crop production, it is essential to balance the chemical elements or nutrients present in the soil with varying parameters of soil like the pH and soil moisture. Crop productivity can be increased to optimum level by efficient soil nutrient management. In case of Nutrient deficiencies, visual symptoms will appear on the leaf. This paper put forwards a method to identify the nutrient deficiencies of plants by making use of visual symptoms appearing on the leaves by Classification. Eight types of deficiencies i.e. N, P, K, Ca, B, Zn and Mg will be studied. The proposed study consists of creation and pre-processing of a set of images consisting of nutrient deficient and healthy leaves, feature extraction and by using Random Forest performing multi class classification of nutrient deficient leaves. Evaluation of tomato leaf from the dataset focuses on recognizing the visual detection and indications of nutritional deficiencies. The proposed architecture achieves the 98.30% accuracy with the model size of 9.20 MB.

Keywords: Balance, Nutritional Deficiency, Visual Symptoms, Classification, Random Forest

I. INTRODUCTION

A. General

In order to increase agricultural production, it is crucial for emerging nations like India to incorporate technical advances in this area. Important research is being done on the issue of qualitative farming, with the goals of raising agricultural production and quality standards, minimizing expenditures, and raising profitability. In crops every nutrient will indicate a deficiency with different separate symptoms. Nutrient reduction depends on the rainfall, fertilizer usage, terrain and the land management practices.

Observable indications of nutrient deficiencies and toxicities would be the -first indication to show that a particular area or location within a field has a nutritional

problem. Identifying and noticing these visual symptoms is very important for taking the corrective action. By efficiently managing the nutrient contents present in the soil deficiencies occurring in plants can be controlled. Additionally, more evaluations can be done with exhaustive leaf and soil sampling. The functions of essential nutrients in plants and its deficiency symptoms are given below. Nitrogen is mainly essential for the building of proteins in plants. It helps in improving the crop yield as well as it is essential for the proper growth of plants. When there is Nitrogen deficiency, colour of leaf turns from green to yellow. The leaf and the veins will become uniformly yellow in colour. In case of acute deficiency, the colour of leaf becomes whitish yellow. Mostly nitrogen is absorbed in the soil as ammonium nitrate [1].

Phosphorous is needed for cell generation and division and in due course results in the growth of plants. Necrosis of older leaves and the undersized plants with short petiole are the notable symptoms. Plants take phosphorous as phosphate from the soil. Phosphorous absorption depends on the pH of the soil. Potassium has crucial role in the plant growth. It is necessary for activating many enzymes during metabolic reactions, controlling the opening and closing of stomata for regulating water loss and for the transportation of sugar. It plays a vital role in photosynthesis as well as carrying the photo assimilates all over the plant. Tip burn of leaves, necrosis between the veins of the leaves and the drooping of leaves and branches are the main symptoms. Potassium is absorbed in the form of ions from the soil. Calcium deficiency results in brown spots of tip burn in young leaves, curling of leaves and diminutive growth of plants.

Boron deficiency causes a reduction in the growth of young leaves, flowers, buds, tip of roots points. Overall growth of the plant will be diminished [2].

Zinc deficiency results in interveinal yellowing of leaves called chlorosis. In cabbage it is exhibited as pale yellow green colour of leaf.

Magnesium is mobile in plants and the symptoms appear initially in the lower leaves. Chlorosis starts from the

margin of the leaf and progresses to the middle and later the colour changes to brownish red. Big veins will be green in colour. Here we provide a method for determining whether or not a cabbage leaf has been infected with a certain blight. Picture recognition is an example of cutting-edge technology that can be used to detect the type of disease that a large crop like cabbage could have. For considering an element as an essential element three points are to be noted:

1. No other element can replace its function
2. The element will be required throughout its life cycle
3. The element will be directly involved with the growth of plant.

As shown in Fig.1, this paper proposes an approach to identify the nutritional deficiencies, utilizing image processing and machine learning approaches. Deficiencies listed above are considered in this paper. The important aspects of this work are:

1. Creating a data set of nutrient deficient leaves and healthy leaves.
2. Examining whether classification based on Random Forest can identify the nutrient deficiency
3. Estimating the performance of this approach.

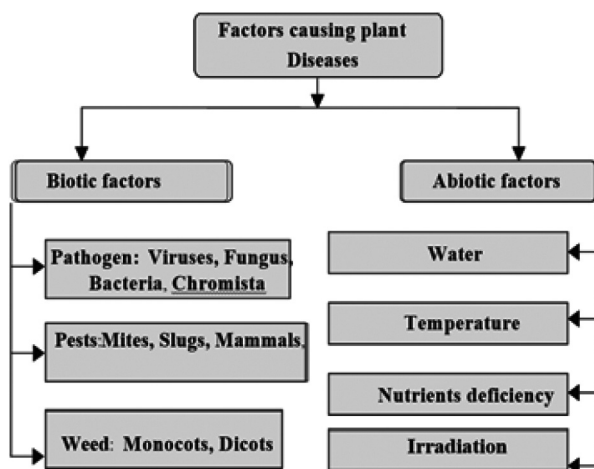


Fig. 1: Factor Causing Plant Disease

II. BACKGROUND AND LITERATURE REVIEW

In this section we compare the many existed work and researches in crops leaf disease. In order to identify plant diseases from photographs of leaves, several authors have created various methods as thresholding, boundary detection, and spot detection algorithms were used to zero in on the sick region of the leaf. Kumar S. Jain et al [3] developed computational intelligent vision sensing system for the estimation of nutrients in plant leaves. Color features of the leaves images taken from the field under different conditions of light are examined. Deep sparse extreme learning machines combined with genetic algorithm is used to normalize plant images. Mean, variance, skewness, and kurtosis are taken as moments of color distribution and nutrients are estimated with these.

Zhiqiang et al [4] determine some steps to improve the soil quality as available nutrients estimation accuracy. Initially the rapid-nutrient assimilation method was used to optimize the crop growth simulation process. Compared with the ensemble Kalman filter. Rapid Nutrient Assimilation method showed a better performance for estimating nitrogen, phosphorus, and potassium

Ukrit [5] used convolutional neural network for identifying plant deficiencies. The network is trained to identify whether the plant shows any symptoms or the deficiency. Image is divided into blocks and given to a set of CNNs. Every CNN is trained for detecting the particular symptoms for that particular deficiency. The responses are combined to one by making use of multilayer perceptron to get the response.

Padmashri et al [6] Authors devised a system by considering maize leaf images deficient in nitrogen, phosphorous and potassium. The data set is prepared with nutrient deficient leaf images. The method was non-invasive way of detecting nutrient deficiency in plants and used Inception V3 CNN.

Sui R et al [7] Identified the nitrogen status in the soil for the cotton crop, by a see., frame work developed in the ground. It consists of optical sensor, an ultrasonic sensor, and a data-capture and processing unit. Nitrogen status was predicted by making use of artificial neural network. Artificial neural network uses the sensor data. Accuracy was ninety percentage. Imran et al [8] used aeroponics in their research. In this plants are grown in growth chamber. Nutrient solution is used instead of soil and nutrient mist is injected at regular intervals. Various parameters like electrical conductivity, carbon dioxide concentration pH of soil etc are measured in regular intervals. Wireless sensors are used for the detection.

Achmad et al [9] used fuzzy logic to give recommendations for sugar cane essential nutrients in the soil. Fuzzification is done on soil moisture, pH, growth phase, and nutrient element.

Arvin et al [10] measured Nitrogen, Potassium and Phosphorus elements present in the soil, moisture and temperature for the growth of paddy by using sensors. Sensor data is analyzed and on the basis of data farmers could choose the suitable rice variant for cultivation.

Goutham K et al [11] conducted their research on paddy crop. The authors measured the nutrients present in soil in real time and send the data to farmers. This helps the farmers to identify the type of paddy crop to be grown for a particular period. Experiments were conducted by measuring nitrogen, phosphorous and potassium in the soil.

The relevant literature survey's indicated that Random Forest could be used for identifying nutritional deficiencies based on the symptoms appearing on leaves.

III. DATA CLASSIFICATION & PROCESSING

A. Acquier Image

Images of plant leaves can be captured either from the field with complex and complicated background [12] or with white background under controlled conditions[13]. Images may be captured through different devices like mobile phones , CCD cameras etc. Quality of image captured is based on the camera and the orientation of the principal axis of lens to the plane of the leaf. Accuracy of nutrient deficiency detection depends on the image capturing process because it is the input for training [18].

Two thousand images of the leaves and plant with a resolution of 3546 X 4608 pixel were taken using Oppo. Three hundred and fifty images of healthy plants and remaining images with nutrient deficiency were acquired.

B. Pre-process The Image

The image captured may contain noise either due to improper brightness, undesirable objects, poor contrast etc. So the image captured will go through noise removal. Acquired image is resized and noise is removed from the image. In median filter, abrupt change in pixel value is replaced with median of the neighboring values of the pixel. On the contrary, bilateral filter removes noise by replacing the abrupt change in intensity value of the pixel by weighted average of the intensity values of neighboring pixel values. In image processing applications Gaussian filter will blur images and removes noise based on Gaussian distribution. Laplacian filters are derivative filters used to detect abrupt change in edges of the image.

C. Feature Extraction

Feature extraction can be considered as a dimensionality reduction. Features can be grouped into shape, texture, colour etc. Length, width, perimeter, area etc. can be considered as features of shape while contrast, homogeneity, correlation, etc. can be considered as features of Texture type.

Nutrient deficient leaves will have brown spots, patches, brown color at the tip, In order to identify these, features should be extracted from the image for that color space of the image may be divided into bins. Each bin correspond to particular color [18]. Pixels are placed in the bin which is very close to its color.

D. Edge Detection

Uniformly green colored leaves are healthy leaves. Edge detection is done to identify the patches and spots. Sobel filter is used for edge detection. In a gray scale image, the sudden change in intensity of the image is treated as edge. Sobel operator makes use of kernel and convolution operation to petrify the edges. Two kernels for approximating the change in intensity values of pixels in x direction and y-direction are used. Approximating the change in intensity values of pixels means approximating the gradient of intensity values. (G_x-horizontally in x-direction

and vertically in y-direction). Magnitude of Gradient for Pixel (x,y)=Sqrt [$G_x^2 + G_y^2$].

For pixel (x,y) direction of gradient $\theta = \tan(G_x / G_y)$.

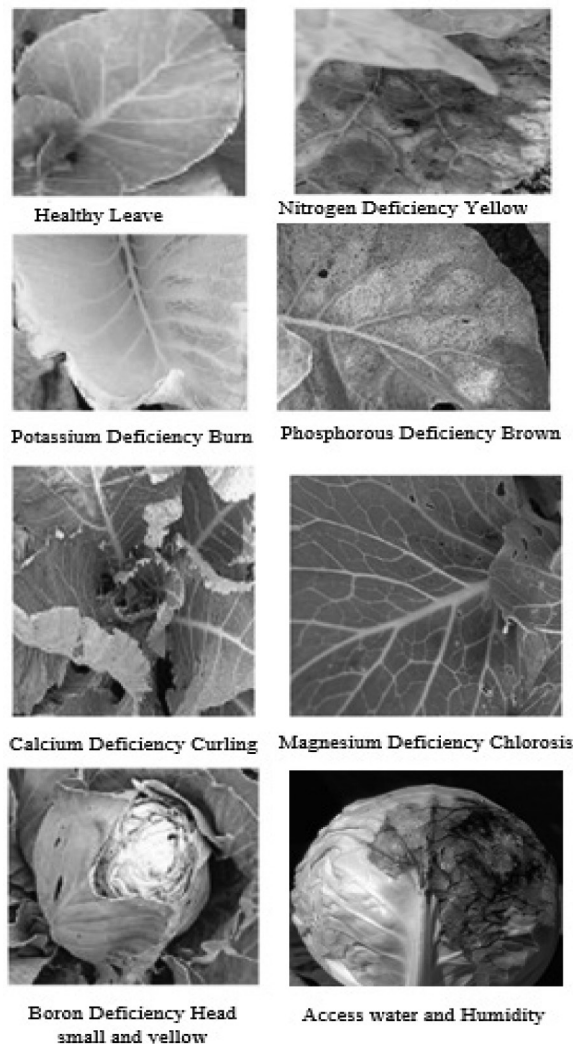


Fig. 2: Nutrient Deficient Leaves

E. Classification

Classification is a supervised learning approach in which the class of new observations or data is identified based on the training data. In this paper Random Forest, an ensemble learning used for classifying the nutrient deficiencies. Random Forest can deal with large datasets and it gives higher accuracy as compared with decision tree.

F. Dataset Used

Initially cabbage was grown hydroponically in completely balanced nutrient solution for 11 weeks. Nitrogen, Potassium, Phosphorous, Calcium, Boron, Zinc and Magnesium deficient solutions and balanced nutrient solution were used for the experiment [22].

Now present research many of datasets are used for testing and validation but the Plant – Village dataset is mostly used due to the large variation of Indian geographic

region. The dataset compares 55,309 images of healthy and infected leaves belonging to crops, labeled by plant pathology experts. Among them, 16,170 images are of Cabbage leaves, which is divided into one healthy and one disease classes. In order to maintain the concentration of nutrient deficient solution, every week the solutions were changed. After 25 days the images of the leaves and plant were taken every day using VIVO V21E. Two thousand leaf and plant images were taken with a resolution of 3546 X 4608 pixel. This dataset offers a wide variety of diseases and contains samples of leaves being infected by various diseases to different extents.

Manually segmented leaves of cabbage are shown in the Fig. 2. Identifying the nutrient deficiency involves and good leaves.

- Leaf or plant analysis of the crop
- Find the correlation of crop yield with any crop related characteristics like the amount of nutrients present in the plant.
- Find Calibrating the correlation result in terms of crop yield
- Interpret the result.

G. Data Processing

In order to develop a model using machine learning three distinct data sets are required.

Training Data

This is the initial data used by the network to learn the hidden parameters automatically.

Test Data

After the model is built using training data set , unseen data is required to test the model, called the test data.

Validation Data Set

There are some hyper parameters which cannot be learned automatically during training. These parameters like learning rate, size of batch needs to be adjusted manually using the validation set.

Images are converted to uniform fixed size. Machine learning requires enormous amount of data in order to get good accuracy for the model. Augmentation creates more images from existing images and it helps in increasing the accuracy of model. In order to identify nutrient deficiency, data annotation of images [15] raises the query to identify the best part of image to send to the network i.e. the entire leaf or part of leaf or the whole plant.

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

A. DL Architecture for Cabbage Leaf Disease Detection

This section outline the research design and the proposed method for identifying and classifying nutritional deficiencies. During this study six nutritional deficiencies-

Nitrogen, Phosphorous, Potassium, Calcium, Boron, Zinc and Magnesium were considered.

This study made use of experimental research design and images of leaves with deficiencies in Nitrogen, Phosphorous, Potassium, and Calcium in Cabbage were considered. Two thousand leaf images of Cabbage were considered. The nutritional deficiencies classified by the model were manually verified by the experts. [20]

Random Forest is an ensemble learning used for classification and regression. Decision trees are created on data models and the nutrient deficiencies are estimated from each of the decision tree. Aggregation is then generated using CART (Classification and Reintegration Tree Algorithm) as shown in Fig. 3 that the Random Forest gives high accuracy as the majority vote form each decision tree is used for classification.

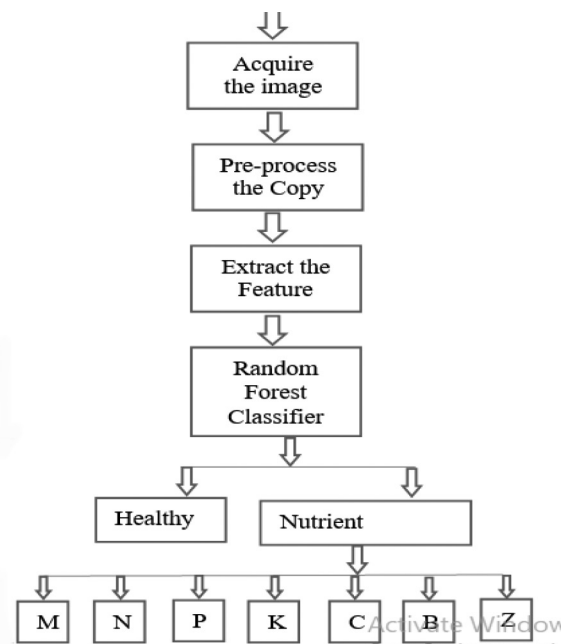


Fig. 3: Proposed System Architecture

B. Classifier Network and Method

Instead of directly using the extracted features from pre trained models for final prediction, we employed a combination of dense, dropout, and batch normalization blocks to fine-tune the extracted traits further. As shown in dense blocks were added before the final output layer. The pre trained MobileNetV2 architecture we used was trained on a large and generalized dataset, making it perfect for feature extraction.

Acquire the images of healthy and nutrient deficient leaves.

1. Preprocess the image
2. Convert the collected images to uniform size. Augment the image to get large set of images by doing different operations on image like rotation, width shift, flip, fill etc.

3. Extract features and annotate the image Labels like colour, spots, tip brown, curly leaf, brown patches, yellow coloured veins are considered.
4. Training data and Test data. Data set was divided into training (80%) and test data (20%).
5. Model building using Random Forest classifier.
6. Classify the leaves into healthy and nutrient deficient leaves.

V. PERFORMANCE EVALUATION

Performance of the system was evaluated using Precision, Recall [16][17] and F-measure [14]. Twenty agriculture experts were also asked to identify the nutritional deficiencies manually from the image set. Prediction of the agriculture experts is then compared with that of the proposed model.

$$\text{precision} = \frac{\text{True_Positive}}{\text{True_Positive} + \text{False_Positive}} \quad (1)$$

$$\text{recall} = \frac{\text{True_Positive}}{\text{True_Positive} + \text{False_Negative}} \quad (2)$$

$$F\text{-measure} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (3)$$

Average of the prediction results of agriculture experts were taken for the calculation of Precision, Recall and F-measure. Using the same metrics the performance of the proposed model was also evaluated. The proposed method gave more accurate deficiency identification as compared with prediction results of agricultural experts.

Counting Accuracy (A) refers to the accuracy of complete counting process.

$$P = 1 - [(\sum m + \sum n)/t] \quad (4)$$

Where

P = Counting Accuracy

m = Over count

n = under count

t = ground truth (actual number of leaves with deficiencies)

TABLE 1: EVALUATION SUMMERY

Class	No of Leaves	Accuracy by Random Forest	Accuracy Agriculture Expert
Healthy Leaf	450	95.8	52
Nitrogen Deficient	325	96.7	48
Phosphorous Deficient	430	94.3	39
Potassium Deficient	325	95.8	35
Calcium Deficient	350	92.8	37
Boron Deficient	230	94.1	40
Magnesium Deficient	320	96.2	36
Zinc Deficient	470	92.3	32

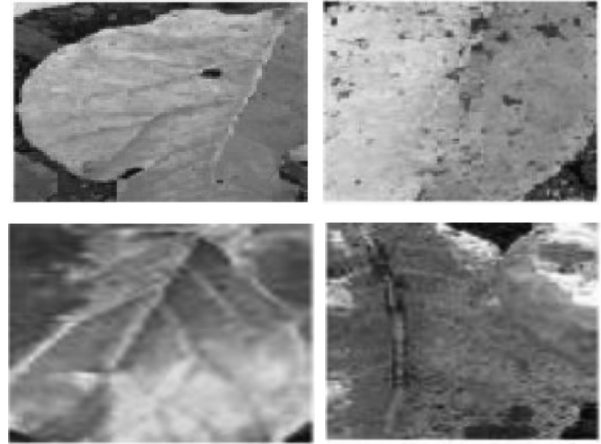


Fig. 4: Variation within the Same Class (Deficiency Nitrogen)

VI. CONCLUSION

The study shows the feasibility of using Random Forest classifier for identifying, nutrient deficiencies present in the cabbage leaves. Random Forest classifier based identification is definitely preferable over identification by human experts as shown in Fig.4. Initial difficulty was that some deficiency symptoms overlap for different nutrients. For example necrosis occurs for Phosphorous as well as for Potassium deficiency. Overlapping symptoms for the same nutrient, progressive appearance of symptoms and variations within the class impose some challenges in the nutrient deficiency identification

The next step is to use the sensor technology for nutrient deficiency identification of the soil. From the sensor signal, it is possible to identify the amount of nutrients present in the soil. Sensor to sensor networking requires routing and synchronization. Physical layer design need to be upgraded for attaining energy saving. Sensing technology helps in enhancing the crop production and results in optimum crop management. In the agricultural field, this will help in attaining sustainability.

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